Haoming Cai

Ph.D. Student, Computer Science, University of Maryland

hmcai@umd.edu
hm-cai.com
Google Scholar

Research Interests

Foundation models for image and video · Physics-aware generation with diffusion models · Computational imaging · Large-scale benchmark design · 3D reconstruction

Education

2022–2027 **University of Maryland**, College Park, MD
Ph.D. in Computer Science. Advisor: Christopher Metzler.
Physics-aware Gen-AI; computational imaging; image signal processing and end-to-end optimization.

2017–2022 **The Chinese University of Hong Kong, Shenzhen**, Shenzhen, China
B.Eng. in Computer Science and Engineering.
Gap year 2020–2021 at XPixel as a research assistant, then part time through 2022 while completing the degree, supervised by Chao Dong.

Research Experience

Summer 2026 **Adobe**, San Jose, CA
Ph.D. Research Intern, NextCam team (Marc Levoy).

- Fast instruction-based local image editing with diffusion transformers: compute scales with the edited region, and everything outside it stays untouched across multi-turn edits. Paper in preparation.

Summer 2025 **Adobe**, San Jose, CA
Ph.D. Research Intern, NextCam team (Marc Levoy), supervised by Shumian Xin and Zhoutong Zhang.

- Turned casual videos into light fields to create physically correct bokeh training data. ECCV 2026; patent filing in process.

Summer 2024 **Dolby Vision Lab**, Sunnyvale, CA
Ph.D. Research Intern, supervised by Guan-Ming Su.

- Portrait lighting control for text-to-image diffusion models without light stage data. Outcome: ICCV 2025, one patent.

2020–2022 **XPixel Group, SIAT**, Shenzhen, China
Undergraduate Research Assistant, supervised by Chao Dong.

- Efficient and controllable image restoration; image quality and aesthetic assessment.
- Outcome: two ECCV papers, two CVPR workshops organized, two Chinese patents (one granted).

Publications

Reverse chronological. * denotes equal contribution. Google Scholar lists citation counts.

ECCV 2026
Homepage
Paper
Large-Scale Light Field Synthesis from Videos Enables Geometrically Consistent Bokeh Editing.
Haoming Cai*, Zhoutong Zhang*, Christopher Metzler, Shumian Xin.

COLM 2026
Homepage
Enhancing Vision Language Models' Robustness Under Adverse Imaging Conditions.
Tianfu Wang, Mingyang Xie, **Haoming Cai**, Tianyi Xiong, Xiyao Wang, Dongdong Fu, Guan-Ming Su, Paola Cascante-Bonilla, Christopher Metzler.

ICCV 2025
Homepage
Parametric Shadow Control for Portrait Generation in Text-to-Image Diffusion Models.
Haoming Cai, Tsung-Wei Huang, Shiv Gehlot, Brandon Y. Feng, Sachin Shah, Guan-Ming Su, Christopher Metzler.

CVPR 2025
Homepage
Repurposing Pre-trained Video Diffusion Models for Event-based Video Interpolation.
Jingxi Chen, Brandon Y. Feng, **Haoming Cai**, Tianfu Wang, Levi Burner, Dehao Yuan, Cornelia Fermüller, Christopher Metzler, Yiannis Aloimonos.

CVPR 2025
Homepage
Flash-Split: 2D Reflection Removal with Flash Cues and Latent Diffusion Separation.
Tianfu Wang*, Mingyang Xie*, **Haoming Cai**, Sachin Shah, Christopher Metzler.

NeurIPS 2024
Temporally Consistent Atmospheric Turbulence Mitigation with Neural Representations.
Haoming Cai*, Jingxi Chen*, Brandon Y. Feng, Weiyun Jiang, Mingyang Xie, Kevin Zhang, Cornelia Fermüller, Yiannis Aloimonos, Ashok Veeraraghavan, Christopher Metzler.

ECCV 2024
Homepage
Flash-Splat: 3D Reflection Removal with Flash Cues and Gaussian Splats.
Mingyang Xie*, **Haoming Cai***, Sachin Shah, Yiran Xu, Brandon Y. Feng, Jia-Bin Huang, Christopher Metzler.

CVPR 2024
CodedEvents: Optimal Point-Spread-Function Engineering for 3D-Tracking with Event Cameras.
Sachin Shah, Matthew Albert Chan, **Haoming Cai**, Jingxi Chen, Sakshum Kulshrestha, Chahat Deep Singh, Yiannis Aloimonos, Christopher Metzler.

NeurIPS 2023
Homepage
Assessor360: Multi-sequence Network for Blind Omnidirectional Image Quality Assessment.
Tianhe Wu, Shuwei Shi, **Haoming Cai**, Mingdeng Cao, Jing Xiao, Yinqiang Zheng, Yujiu Yang.

ICCV 2023
Homepage
Snow Removal in Video: A New Dataset and A Novel Method.
Haoyu Chen, Jingjing Ren, Jinjin Gu, Hongtao Wu, Xuequan Lu, **Haoming Cai**, Lei Zhu.

ECCV 2022
Homepage
Super-Resolution by Predicting Offsets: An Ultra-Efficient Super-Resolution Network for Rasterized Images.
Jinjin Gu, **Haoming Cai**, Chenyu Dong, Ruofan Zhang, Yulun Zhang, Wenming Yang, Chun Yuan.

ECCVW 2022
Homepage
Efficient Image Super-Resolution Using Vast-Receptive-Field Attention.
Haoming Cai*, Lin Zhou*, Jinjin Gu, Zheyuan Li, Yingqi Liu, Xiangyu Chen, Yu Qiao, Chao Dong.

CVPRW 2022
Homepage
Blueprint Separable Residual Network for Efficient Image Super-Resolution.
Zheyuan Li, Yingqi Liu, Xiangyu Chen, **Haoming Cai**, Jinjin Gu, Yu Qiao, Chao Dong.

CVPRW 2021
Homepage
Toward Interactive Modulation for Photo-Realistic Image Restoration.
Haoming Cai, Jingwen He, Yu Qiao, Chao Dong.

ECCV 2020
Homepage
PIPAL: A Large-Scale Image Quality Assessment Dataset for Perceptual Image Restoration.
Jinjin Gu, **Haoming Cai**, Haoyu Chen, Xiaoxing Ye, Jimmy S. Ren, Chao Dong.

Preprints and Manuscripts

arXiv 2026
Homepage
Real2SAM2Real: Generative 3D Caches as Complementary Context for Video Diffusion.
Jiayi Wu*, **Haoming Cai***, Cornelia Fermüller, Christopher Metzler, Yiannis Aloimonos.

In preparation Precise and Fast Local Image Editing.

In preparation What If? Guided Diffusion as a Tool for Scientific Inquiry into Tropical Cyclones.

Patents

2026 Lighting customization of text-to-image diffusion models. EP 4760643 A1, pending. Dolby; first inventor. Google Patents

2025 Image super-resolution method based on blueprint separable residual network. CN 115082306 B, granted. SIAT; English title translated. Google Patents

2023 Image processing method, apparatus, electronic device and storage medium. CN 115797165 A, pending. SIAT; English title translated. Google Patents

Academic Service

2021 & 2022 **Co-organizer**, NTIRE Challenge on Perceptual Image Quality Assessment, CVPR Workshop. Jinjin Gu, **Haoming Cai**, Chao Dong, Jimmy S. Ren, Yu Qiao, Shuhang Gu, Radu Timofte.

- Addressed generative distortion, a failure mode existing IQA metrics did not capture.
- Methods introduced through the challenge include IQT (150+ citations) in 2021, and MANIQA (740+) and AHIQ (130+) in 2022.
- Benchmark integrated into IQA toolboxes including *PyTorch-Image-Quality* and *IQA-PyTorch* (4,700+ combined GitHub stars); 800+ participants.
- NTIRE 2021 · NTIRE 2022

Reviewer CVPR (2022–2026) · ECCV (2022, 2024, 2026) · NeurIPS (2024–2025) · ICCV (2025) · ICLR (2025) · ICML (2025) · AAAI (2023–2024) · WACV (2023, 2025) · TMM · TMLR

Honors and Awards

2026 Google PhD Fellowship Nominee, nominated by the University of Maryland.

2025 CVPR Outstanding Reviewer. Link

2024 Finalist, Qualcomm Innovation Fellowship (18% acceptance). Link

2022–2024 Ph.D. Dean Fellowship Award. Graduate School, University of Maryland.

2022 Winner, Efficient Image Super-Resolution Challenge, model complexity track. NTIRE Workshop, CVPR 2022.

Technical Skills

Languages Python · C++ · CUDA · PyTorch · NumPy · OpenCV · Git

Methods Machine learning · Deep learning · Diffusion models · Vision-language models · 3D reconstruction and Gaussian splatting · Neural representations · Self-supervised learning · Model efficiency

Domains Computational photography · Computational imaging · Image signal processing (ISP) · HDR · Event cameras · Image quality assessment